

From Your Pocket to Your Boss's: Cash Transfers & Wage Theft

Taylor Watson *

September 29, 2023

Abstract

Wage theft is a pervasive issue for American workers, yet research on the subject is stymied by underreporting: workers fear retaliatory firing if they report. This paper shows that worker reporting decisions are affected by work-contingent government transfers, as they make remaining employed more attractive, increasing the cost of being fired. Using variation in state Earned Income Tax Credits over time, I find that increases in transfers are associated with lower reporting counts, analyzing at the ZIP code level. To confirm that this decrease truly reflects changes in reporting rather than a decrease in wage theft itself, I estimate the relationship between transfers and several intensive-margin outcomes of wage theft. Finding no evidence that transfers decrease wage theft intensity, I conclude that transfers reduce reporting propensities.

*Boston University, watson@bu.edu

I Introduction

Wage theft is a pervasive issue for American workers, wherein employers avoid paying workers their legally due wages. This can take many forms, such as failing to meet the minimum wage or withholding overtime pay. While it is known to be a widespread issue (costing workers an estimated \$50 billion annually) it is not well-studied due to underreporting — workers who report wage theft are often fired in retaliation, creating a significant deterrent (Meixell and Eisenbrey 2014).

To better understand how workers make reporting decisions, I explore the role of work-contingent government transfers. By increasing a worker’s total income, these transfers should increase the value of work and make potential retaliatory firing even more costly. To study this relationship, I propose a model in which workers weigh the potential gains from reporting (i.e., backwages repaid) against the potential costs (i.e., retaliatory firing). Simple comparative statics suggest that an increase in government transfers should decrease reporting propensities, as the loss from being fired outweighs the potential gains from a successful report.

Taking this model to the data on wage theft (documented by the Federal Wage & Hour Division) this prediction is borne out. As a measure of work-contingent transfers, I use differences in Earned Income Tax Credit (EITC) matching generosity. States match the federal EITC at varying rates, and many have adjusted those rates over time, producing plausibly exogenous variation at the state-year. Analyzing this relationship at the ZIP-code level via Poisson regression, I find that an increase in state EITC of 10 percentage points is associated with a decrease in ZIP wage theft reports of approximately 4.2%. Scaling that up to the state level, this implies roughly 26 fewer reports per year (off an average of 600) and an additional \$286,000 in unreported stolen wages.

Wage theft is a complex challenge for American workers. Most estimates of its scale rely on small surveys, and what is formally documented only reflects those cases where the victims have chosen to come forward (Cooper and Kroeger 2017). It is particularly common in industries with more complex wage structures, such as tips or productivity bonuses, but

has been perpetrated by many kinds of employers, from large firms like Amazon or Bank of America to small businesses with just two employees (Sainato 2023). The issue has been of accordingly high interest in the economics literature, including studies of what factors make firms more likely to engage in theft (Ji and Weil 2010; Clemens 2021) and how the government can best deter this behavior (Weil 2010, 2018; Stansbury 2021); however, there remains little study of how workers themselves respond to wage theft.

When workers are victims of wage theft, their primary recourse is to file a report to the Wage & Hour Division of the Department of Labor.¹ This will usually lead to an investigation by the Division into what labor laws have been violated, and what backwages (or possible fines or damages) are owed. By one estimate, roughly \$1 billion is recovered each year - accounting for a minute fraction of the estimated theft, as the vast majority of victims do not report the crime (Meixell and Eisenbrey 2014).

Unfortunately, due to this underreporting there is still much that is not understood about worker wage theft behavior, in particular as it interfaces with the welfare state and government transfers. The crime of wage theft falls most heavily on workers in low-wage industries who are also likely to rely on welfare and transfer programs, making the interaction of the two important for optimizing policy. The lack of research in this area was highlighted by a recent review article, Marinescu and Rosenfeld (2022), in the context of diminishing worker power.

This paper begins to fill this gap in the wage theft literature, while also contributing to the growing literature on workplace crime and worker reporting incentives. As noted above, the observed data on various employer violations depend jointly on the true intensity of those violations and on the likelihood that violations are converted into documented reports (Marinescu et al. 2020). The importance of this distinction was made by companion papers Boone and Ours (2006) and Boone et al. (2011) in the context of workplace safety. While Boone and Ours (2006) show that empirically it appears workplace safety violations fall

¹In most states, workers also have the opportunity to file a complaint with their state agency; however, the data on these state-level reports is heterogeneous and ill-maintained, making analysis of it highly challenging. Workers could also sue their employer directly in court, though this is often expensive and a lengthy process.

during recessions, Boone et al. (2011) clarified that the effect is largely driven by changes in reporting behavior, rather than actual workplace safety changes.

Boone et al., offer a clear mechanism for the effect of recessionary shocks on reporting. The primary deterrent for workers considering reporting is the threat of retaliatory firing. The decision to report or not must weigh the potential costs of being fired and becoming unemployed against remaining quiet and staying employed. This comparison of unemployment conditions to continued employment naturally depends on the outside unemployment rate. For example, as a recession decreases the outside option of workers if they become unemployed, the risk of a retaliatory firing for reporting workplace safety violations grows more costly, and thus reports fall. A similar mechanism has been documented in a variety of different contexts: Hispanic workers facing increased threats of immigration enforcement report fewer safety violations (Grittner and Johnson 2022), low-income workers with fewer outside options are less comfortable discussing workplace issues (Hertel-Fernandez 2020), and reductions in unemployment benefits exacerbate underreporting of sexual harassment in the workplace (Dahl and Knepper 2021).

Across these settings, authors focus on the worsening unemployment situation as the identifying variation for the mechanism. In my paper, I employ a separate but similar logic, that an improvement in the condition of remaining employed should also decrease reporting, as the cost of becoming unemployed increases for the worker. Using increases in the work-contingent transfers as an increase in the attractiveness of employment. While I focus on the crime of wage theft, a similar complementary mechanism likely exists for the other kinds of workplace crime which are studied in the literature.

While I emphasize the effect of the state EITC matching rate, it is more accurate to consider the effect of both the EITC matching rate and the local EITC participation rate jointly. In my proposed model, I focus on a representative worker and thus eschew the role of the participation rate, but since my empirical work is at the ZIP code level, the participation rate must be accounted for. The participation rate modulates the mechanism of effect through which the matching rate operates - if the participation rate was zero, there

should be no effect of changing the matching rate on wage theft. I therefore extend my analysis to this interaction, and find similar results: the interaction between matching and participation rates has a significant negative effect on wage theft reporting. Additionally, the matching rate alone has a nearly null effect (as expected, as varying this in the context of zero participation should not matter), while the participation rate alone has a significant negative effect. This likely reflects the role of the federal EITC, which allows increases in the local participation rate to decrease wage theft reports, by a similar mechanism to described for the state matching rate.

While I find a negative relationship between transfers and wage theft reporting across several empirical specifications, it is not clear *a priori* that this finding reflects the mechanism proposed. Reported wage theft is a product of both the propensity of workers to report and the propensities of employers to engage in the underlying wage theft. It could be the case that changes in EITC rates are actually affecting this second factor, the intensity of wage theft. This would result in a lower reporting frequency mechanically, as there is less real theft for workers to report. In other words, the negative relationship between transfers and observed reports could be driven by a mechanism of worker behavior (lowering their reporting propensities), or through a mechanism of employer behavior (lowering their wage theft propensities).

To understand this alternative mechanism, one can imagine an employer who sees the increased transfer payments as insulating their employee against the costs of being unemployed, thus diminishing the potency of the employer's retaliatory firing threat. Such an employer would likely decrease their wage theft behavior in response to rising transfers, in order to forestall reports from their newly flush workers, which could be costly to the employer.

In order to differentiate between these two mechanisms, worker and employer behavior changes, I analyze the effect of transfers on secondary outcomes which proxy for employer behavior. Going beyond the reporting frequency itself, these measures tell us something about the nature of the wage theft being reported. Essentially, if the observed decrease

in reporting is being driven by employer behavior, one would expect that these secondary measures would also show some adjustment. Specifically, I use three metrics of employer wage theft: the number of "empty" reports (where no theft or violations are found), the number of legal violations found during investigations of reports, and the amount of civil monetary penalties assessed per report.

Across these outcomes, I fail to find evidence of reduced wage theft intensity or changes in behavior on the part of employers. While individually each of these secondary outcomes is not totally compelling, in combination they are convincing that the decrease in reporting is not driven by the employer mechanism, but rather via my hypothesized mechanism of decreased worker reporting propensities.

II Theory

To understand why wage theft is often not reported by its victims, it is helpful to consider the problem through the lens of a simple model. This section walks through one such framework² which predicts that increased transfer generosity will decrease reporting probabilities. The intuition is that transfers increase the value of employment relative to unemployment, such that the threat of being fired becomes sufficiently costly to outweigh the potential gains from reporting.

Consider an environment with a representative worker facing some exogenous amount of wage theft, $t \geq 0$. I abstract from the employer side of this relationship for simplicity. The worker, faced with a fixed wage w and theft t , must choose whether or not to report that theft to the authorities.

If the worker chooses not to report the theft, they will receive (with certainty) their wage net of theft and net of taxes τ , represented as $(1 + \tau)(w - t)$. Note that in this setup τ will be assumed positive, reflecting a situation similar to the real-world context of the Earned Income Tax Credit. This is appropriate given our understanding of wage theft as a primarily low-wage phenomenon. For the predictions of this model, it is not important that τ be positive, only that the changes in tax structure result in increased net income for the worker.

If the worker chooses to report the theft, their outcome is less certain. A report will trigger an investigation into the situation by the government, which will succeed with a exogenous probability p . This uncertainty reflects the administrative challenges in proving wage theft, as employers may be able to evade detection. A successful report results in repayment of backwages, equal to the amount taken t , plus potentially additional damages, represented as γ . This income would also be subject to taxes. An unsuccessful report returns nothing for the worker.

²This model is based off that of Boone and Ours (2006) for the similar reporting issue of workplace safety, which has also been similarly applied in Dahl and Knepper (2021) for sexual assault reporting. My modification is to introduce tax rates.

Simultaneously, a report creates the risk of a retaliatory firing by the firm, occurring with probability θ . A fired worker will either find a replacement job immediately (at new wage level w'), or they will remain unemployed and earn welfare benefits b . These events occur with probability q and $1 - q$, respectively. Accordingly, the value for the worker of not being fired, V_{NF} is their net wage per the success of their report (and net of taxes), $(1 + \tau)(w - (1 - p)t)$. The value for the worker of being fired, V_F is the probability-weighted sum of re-employment and unemployment, or $q(1 + \tau)w' + (1 - q)b$.

Formalizing this setup, we can calculate the expected value for the worker of making a report, and consider when it may outweigh the welfare value of not making a report:

$$(1 + \tau)p\gamma + \theta V_F + (1 - \theta)V_{NF} \geq (1 + \tau)(w - t) \quad (1)$$

The first term represents the potential payout of the report, the next two terms the uncertainty of potential retaliatory firing, and the right-hand side the status quo. This can be rewritten to compare the expected gains from reporting on one side to the expected losses on the other:

$$(1 + \tau)p\gamma + (1 - \theta)(1 + \tau)pt \geq \theta[(1 + \tau)(w - t) - q(1 + \tau)w' - (1 - q)b] \quad (2)$$

With this simplification, note that the gains on the left-hand side are increasing with respect to the amount of theft t , which is to say that the potential gains from choosing to report are increasing in theft. In contrast, the right-hand side is decreasing in t , as greater wage theft reduces the take home pay of the worker, and thus their potential losses from remaining employed and silent. Combining these two properties, it is clear that there must also therefore exist a level of theft, t^* such that this condition holds with equality. Thus, there exists a tipping point degree of wage theft which can induce the worker to report or not report accordingly. Formally, there exists some t^* such that:

$$(1 + \tau)p\gamma + (1 - \theta)(1 + \tau)pt = \theta[(1 + \tau)(w - t) - q(1 + \tau)w' - (1 - q)b] \quad (3)$$

which can therefore be solved for the expression of that level t^* :

$$t^* = \frac{\theta[(1 + \tau)w - q(1 + \tau)w' - (1 - q)b] - (1 + \tau)p\gamma}{(1 + \tau)[(1 - \theta)p + \theta]} \quad (4)$$

This value is the indifference point for the worker: at a given level of wage theft, t , the decision to report boils down to whether that theft is above or below this indifference value. If $t > t^*$, the worker should report, and if $t < t^*$, they should not.

To understand how changes in the tax structure may affect reporting, consider now a mass of identical workers at identical wages facing a distribution of theft levels. If t^* rises, then some fraction of those workers will be pushed from a position where they prefer reporting into a position where they prefer not reporting, as their respective t falls below the threshold. In other words, a rise in t^* predicts a fall in reporting.

To assess whether taxes, τ , in this model will increase t^* , the derivative of the previous expression can be taken with respect to τ . While the algebra is tedious, the sign of the resulting expression can be condensed to the sign of the following term:

$$(1 + \tau)p\gamma + \theta(1 - q)b \quad (5)$$

This expression must always be positive given the parameters involved, thus predicting that an increase in the tax transfer will increase the critical value t^* . Accordingly, this suggests that an increase in transfers should decrease the propensity of workers to report.

This is the key prediction of the model which will be tested in this paper: an increase in cash transfers via the EITC should be associated with a decrease in reporting of wage theft.

III Data

This analysis requires three data components: (1) wage theft reports, (2) state EITC rates, and (3) relevant control variables. This section summarizes the sources used for these data, how they were assembled into final sample, and presents summary statistics of that sample in Table 1.

III.A Wage Theft Data

The most-used archive of wage theft reports comes from the US Wage & Hour Division, who maintain thorough documentation of each complaint they receive. Their data extends back to 2005, and includes several variables required for this analysis: address of each complaint, the amounts claimed and awarded, and more. While this data is quite rich, there are two challenges with using this data to measure the desired outcome of report frequency.

Firstly, this dataset contains two kinds of cases: cases where an employee complained to the government for assistance (worker-initiated case), and cases where the federal government proactively investigated a firm for potential wage theft (agency-initiated case). Only the first case type is interesting for the purposes of this analysis, as an instance of a worker reporting wage theft, while the latter case type is unhelpful. The two case types are not distinguished in the data, in order to protect the privacy of the complainee. From the perspective of my analysis, this creates an overreporting bias in the count data, a form of nonclassical measurement error.

In the current paper I will take this mixed data as given, treating all cases as worker-initiated reports.³ I offer a robustness test in Appendix ??, which attempts to overcome the issue through a restriction on the analysis sample. Using what is known about the DoL’s targeting strategy for their agency-initiated cases, I try to limit my sample to the cases which are most likely to be worker-initiated. The DoL consistently targets their investigations to locations where there is known to be high wage theft, to industries with high prevalence

³Separately, I am working to more accurately separate these two case types through a generalized Poisson-Poisson mixture model which estimates separate counts for worker-initiated and agency-initiated reports, as demonstrated in other over-reporting contexts by Pararai and Famoye (2022).

of wage theft, and to areas where they believe their investigation will be able to create strong local deterrence effects (Weil 2010). Accordingly, I drop from my sample the top 0.25% of ZIP codes by wage theft report count.⁴ These ZIP codes account for roughly 17% of all wage theft reports in my sample — for comparison, the share of all cases in the data which are agency-initiated is roughly 25% per year, meaning that this data restriction likely drops primarily agency-initiated cases. Repeating my analysis on this restricted sample returns similar results, suggesting that my findings are not being driven spuriously by agency-initiated cases.

The second issue with the Wage & Hour dataset is that it is only a subset of all wage theft reports. Workers in the US have two avenues by which to seek redress of wage grievances: the federal DoL and their local state agency. Thus, not only does the federal file measure just a subset of all reports, it is likely a selected sample: individuals in states with effective local complaint management may be more likely to report there than in states with less effective local management, who may prefer the federal option. Unfortunately, collection of individual state wage theft archives is impossible due to the differences in data management systems across states.⁵

Variation in state agency engagement would present a potential source of bias if there was strong correlation between agency engagement and state EITC rates. For example, if states with generous EITC matching also had strong state agency enforcement, this could create a negative correlation between state EITC and federal wage theft reports. This is unlikely to be the case. Firstly, state EITC rates vary across years within states frequently, creating variation within the relatively unchanging state agency enforcement context, as state wage and hour laws are infrequently changed. Secondly, a study of state-level policies specifically found that many policy changes taken by states in recent years have had an insignificant effect on wage theft reporting (Galvin 2016). To attempt and address any

⁴Specifically, I drop all ZIP codes which experience greater than 10 wage theft cases in any year. These are all ZIPs which regularly experience high levels of cases, and do not appear to be "one-off" instances which otherwise experience low wage theft.

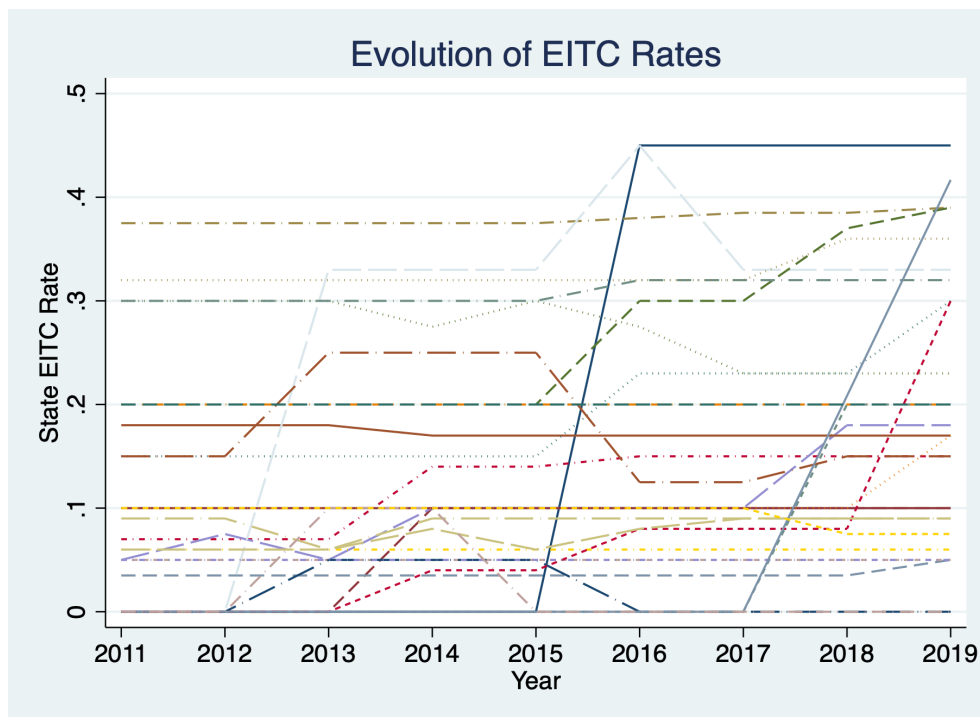
⁵I have begun collecting as much data as is feasible from individual states, and am separately working on a cohesive dataset from this work, but it is limited in scope across only roughly 20 states, and only the past three years for many of those states.

remaining potential bias from this source, I include state fixed effects in my analysis and control for the political party of the state governor in each year (as they have the greatest control over the state agencies enforcement activity and intensity). My results are therefore unlikely to be significantly altered by bias stemming from this data construction issue.

III.B State EITC Data

The explanatory variable of interest, state EITC matching rates, were sourced from the Tax Policy Center at the Urban Institute & Brookings Institution (Tax Policy Center 2023), with gaps filled in using the NBER TAXSIM calculator (Shapiro 2019). In most states, a single matching rate is provided, meaning that the state matches that percentage of the federal EITC; however, in some states there are different rates depending on factors like number of children. These nuances necessitate some interpretive choices, which I explain in more detail in Appendix B. The evolution of rates under my interpretation are summarized in Figure 1 (excluding those states with no matching at any point for simplicity).

Figure 1: Evolution of Rates



III.C Control Data

With the outcome variable and explanatory variable in hand, I then incorporated a number of relevant control variables

Tax Return Statistics

Detailed information on the tax returns filed in each ZIP code are available at the IRS Statistics of Income archive (Internal Revenue Service 2022). In particular, I pulled the total number of tax returns and the share of returns which claimed an EITC benefit for inclusion in my analysis. This is what I'll use as a measure of the local EITC participation rate, particularly in the interaction analysis in Section V.

Business Patterns

The Census County Business Patterns archive includes detailed data at the ZIP code level (U.S. Census 2023b). This includes the number of establishments, the employment share of firms with 50 employees or less, and the share of establishments in several key wage-theft-prone industries: agriculture, construction, and food services.

Demographics

Demographic control variables were taken from the American Community Survey Five-Year Data, including variables for race, education, nativity, poverty, and public assistance (U.S. Census 2023a). There are two variables which were not available from the ACS, namely the population density and unemployment rates at the ZIP level. These are available at five-year frequency from the National Neighborhood Data Archive, and I use them as follows: the estimate of 2008-2012 applied over those years, for 2013-2015 applied to those years, and for 2016-2020 applied to those years (Melendez et al. 2023).

State-Level Variables

Three key variables were not available at the ZIP code level, so I used the available state-year level data. State minimum wages in 2020 dollar equivalents were sourced from the Wage

& Hour Division (US Department of Labor Wage & Hour Division 2023).⁶ Unionization rates were sourced from Unionstats (Macpherson and Hirsch 2023). The party of state governors was sourced from the ICPSR (Kaplan 2023) and validated via the National Governor’s Association.

III.D Sample Construction

The various datasets were merged on the ZIP-code year basis, then the sample was trimmed in various ways to avoid outliers driving the results. Firstly, I restricted to those ZIP codes which correspond to the fifty US states, and exclude other ZIP codes for territories, Washington DC, or other administrative units. Second, ZIP codes which at any point reported fewer than 250 tax returns were dropped across all years, as tax information was not reported for any ZIP below this threshold, and is essential to the analysis.⁷ Third, I winsorize the main outcomes at the 1% and 99% levels, to reduce the influence of outliers.

Temporally I restrict my analysis to run from 2011-2019, the former being the first year many of the relevant control variables are available, and the latter cutoff allowing me to avoid the years of the COVID-19 pandemic. These years display radically different wage theft behaviors, and are subject to a host of other concerns that they may otherwise be different in unaccountable ways.

The final sample contains 25,339 ZIP codes, most of which are observed in all nine years 2011-2019 (for a total of 218,947 ZIP-year observations). This sample and the variables used in my analysis are summarized in Table 1.

⁶My thanks to Joe Lisle who makes these data more easily available through a compilation available here: <https://github.com/Lislejoem/Minimum-Wage-by-State-1968-to-2020/tree/master>.

⁷These dropped zip codes account for roughly 2% of all wage theft cases in the data, making their removal unlikely to drive the results.

Table 1: Summary Statistics

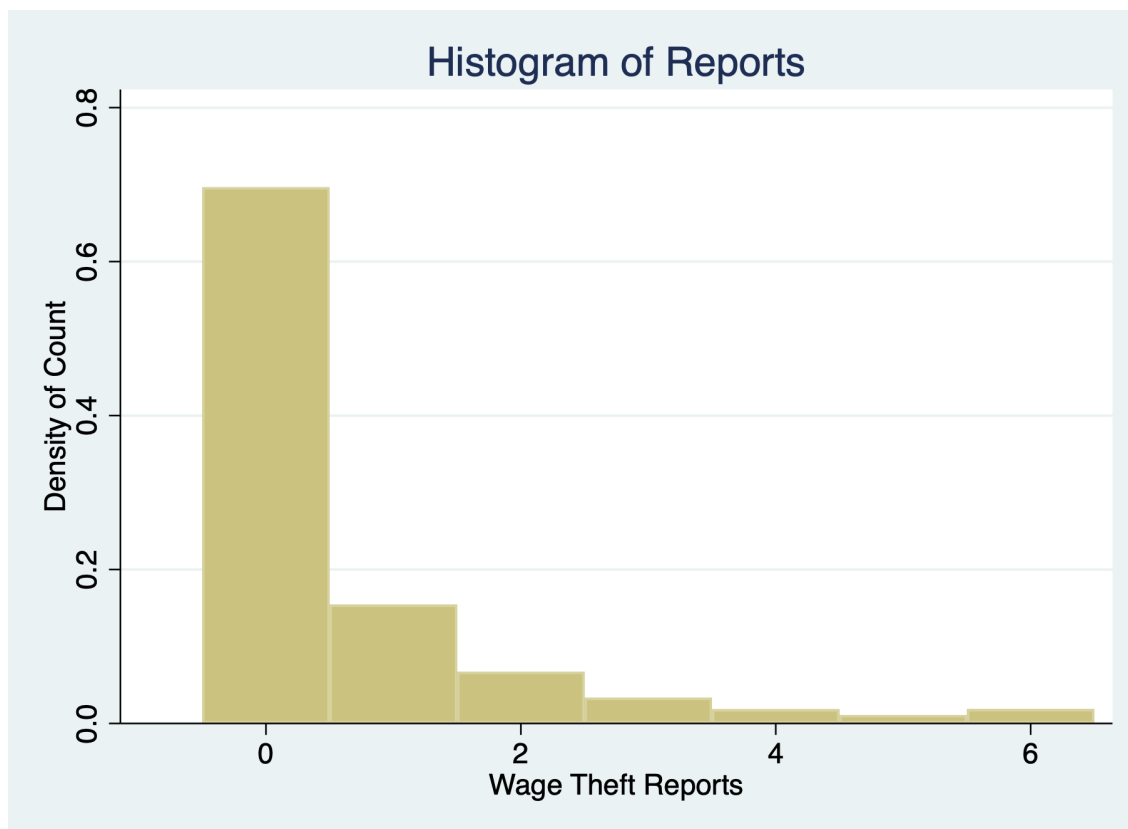
	Mean	Std. Dev.	Min	Max
Wage Theft Reports	0.63	1.25	0.00	6.00
State EITC Rate	0.10	0.16	0.00	0.85
State Unionization Rate	0.12	0.05	0.03	0.26
State Minimum Wage	8.37	1.48	2.02	13.66
Tax Returns	5,862	7,066	250	59,000
EITC Participation Rate	0.33	0.12	0.00	0.84
Share Below Poverty	0.10	0.08	0.00	1.00
Share Black	0.09	0.16	0.00	1.00
Share <HS Education	0.13	0.09	0.00	0.92
Share Noncitizen	0.04	0.06	0.00	0.61
Share Receiving Public Assistance	0.25	0.17	0.00	1.00
Share Establishments in Construction	0.13	0.10	0.00	1.00
Share Establishments in Agriculture	0.01	0.03	0.00	1.00
Share Establishments in Food Services	0.08	0.06	0.00	1.00
Share Employment in <50 Worker Establishments	0.97	0.03	0.00	1.00
Population Density	1,545	5,478	0.05	159,897
Unemployment Rate	0.05	0.03	0.00	0.74

IV Reporting Analysis

IV.A Reporting - Empirical Approach

Given the nature of the outcome of interest (the number of wage theft reports in a given ZIP code-year) it is appropriate to employ a count regression analysis. The distribution of report counts across ZIP-years (seen below in Figure 2) is positively skewed with a large proportion of zeroes - I will accordingly use a Poisson regression as my preferred specification. Note that while just over 71% of observations have no reports, less than a quarter of ZIP codes never experience a report in any year of the observation period.

Figure 2: Histogram of Reports



Even with the correct regression specification, the relationship between state EITC rates and report frequencies may be corrupted by endogeneity, by primarily omitted variables and reverse causality. Addressing these in turn, I believe that my set of covariates adequately control for the relevant variables. The relative sizes of ZIP codes are accounted for through

the number of establishments and the number of tax returns filed. The labor market conditions are measured through the ZIP unemployment rate, poverty level, public assistance engagement, and the state unionization rate. Furthermore, the industrial makeup of the ZIP is controlled for through the establishment shares of key wage theft industries (agriculture, construction, food service) and the share of employment at smaller (<50 worker) firms. On top of these detailed controls, I account for unobserved differences in state environments through state fixed effects, as well as through national shocks through year effects. I believe this comprehensive set of controls is sufficient to ensure that the estimated effect of state EITC rates is free of contamination from omitted variables.

With regards to reverse causality, it seems highly unlikely that the setting of state EITC generosity is done in consideration of the frequency of wage theft in that state's ZIP codes. These rates are not only set at the state level, far removed from any given ZIP code, but are almost always considered in the context of broader tax reforms, which are subject to numerous higher-order concerns such as fairness, budgetary needs, and economic conditions. That the occurrence of wage theft at a particular level would be a contributing factor to these decisions seems implausible.

To estimate the effect of state EITC rates via maximum likelihood, I assume the following conditional fixed effects Poisson model:

$$E[R_{zy}|\delta_s, \gamma_y, SEITC_{s(z)y}, X_{zy}] = \exp(\delta_s + \gamma_y + \beta \cdot SEITC_{s(z)y} + \Theta X_{zy}) \quad (6)$$

where R_{zy} is the number of wage theft reports in ZIP code z in year y , δ_s are state fixed effects, γ_y are year effects, X_{yz} are the various control variables mentioned, and β is the coefficient of interest on the variable $SEITC_{s(z)y}$, the state EITC rate for the state s of ZIP code z in year y . Estimation of Equation 6 by quasi-maximum likelihood with fixed effects leads to consistent estimates (Wooldridge 1999).

IV.B Reporting - Results

Poisson modelling of wage theft reporting is summarized in Table 2. Beginning with a naive regression using the single explanatory variable of state EITC rate, Column 1 estimates a statistically significant negative effect, with a coefficient of -1.311. The sign and significance of this estimate is robust to the inclusion of two-way fixed effects for state and year in Column 2 as well as a battery of controls in Column 3 (though the magnitude falls to -0.402) . To further assess the strength of this result, Column 4 uses ZIP code fixed effects in place of state fixed effects, controlling for greater unobserved heterogeneity, producing highly similar results (coefficient -0.361). Finally, Column 5 estimates the effect in a binary probit model, where the outcome is whether any wage theft reports occurred in a given ZIP-year. Here as well I find a significant negative effect.⁸

While not the focus of this analysis, it is worth noting the coefficient on the ZIP code unemployment rate, for comparison with previous work like Dahl and Knepper (2021) and Boone et al. (2011). They use unemployment conditions as the main variation of interest in explaining changes in reporting behavior, finding that worsening unemployment conditions have a suppressive effect on reporting of workplace crime. In line with those papers, I find that an increase in the unemployment rate has a negative correlation with reporting, with a sizeable effect estimate in Columns 3 and 5.⁹

⁸While I have clustered my standard errors at the level of the state-year to allow for correlation of the errors within those units, this in turn creates an issue of heterogeneously sized clusters. I address concerns related to this issue in Appendix Section C. In brief, the results presented here are robust to bootstrapping inference through the wild score cluster bootstrap, as recommended by MacKinnon et al. (2023).

⁹While the effect becomes insignificant in Column 4 with ZIP code fixed effects, it seems likely that here the ZIP effects are absorbing significant unobserved heterogeneity related to labor market conditions with a similar effect.

Table 2: Wage Theft Report Results

	(1)	(2)	(3)	(4)	(5)
	Naive Poisson	TWFE Poisson	Complete Poisson	ZIP FE Poisson	Probit
State EITC Rate	-1.311*** (0.283)	-0.367*** (0.120)	-0.402*** (0.136)	-0.361*** (0.133)	-0.203** (0.102)
State Unionization Rate			-0.680 (0.856)	-0.577 (0.833)	-0.828 (0.660)
State Minimum Wage			0.004 (0.015)	-0.000 (0.014)	-0.011 (0.010)
Governor Party			0.011 (0.026)	0.010 (0.025)	-0.005 (0.020)
Log Tax Returns			0.861*** (0.009)	0.194*** (0.070)	0.635*** (0.005)
EITC Participation Rate			-0.908*** (0.156)	0.280 (0.209)	-0.337*** (0.096)
Share Below Poverty			1.523*** (0.105)	0.306** (0.136)	0.961*** (0.088)
Share Black			-0.117* (0.068)	0.390** (0.184)	0.114** (0.050)
Share <HS Education			-0.431*** (0.108)	-0.167 (0.188)	-0.360*** (0.092)
Share Noncitizen			0.866*** (0.152)	-0.078 (0.245)	1.618*** (0.130)
Share on Pub Assist.			0.537*** (0.064)	0.037 (0.069)	0.125*** (0.044)
Share Est in Cons			-2.348*** (0.123)	0.478* (0.289)	-0.917*** (0.066)
Share Est in Ag			-1.438*** (0.404)	0.297 (0.758)	-0.055 (0.168)
Share Est in Food Svc.			0.592*** (0.177)	0.580** (0.294)	0.900*** (0.101)
Share Emp <50			-7.550*** (0.559)	-1.358*** (0.355)	-5.556*** (0.208)
Log Population Density			-0.012* (0.007)	0.040 (0.031)	-0.002 (0.005)
Unemployment Rate			-1.413*** (0.348)	-0.017 (0.282)	-0.809*** (0.217)
State & Year FE		✓	✓	ZIP & Year	✓
R ²	0.00	0.06	0.32	0.31	0.31
N	218,930	218,930	218,930	145,975	218,930

Note: Standard errors are clustered at the state-year level. * = $p < 0.1$, ** = $p < 0.05$, *** = $p < 0.01$.

V Interaction Analysis

Thus far I have analyzed the effect of state EITC matching rate on wage theft reports; however, this analysis has abstracted away the interaction between EITC matching rates and local EITC participation rates. There is well-known heterogeneity in participation across (and within) states, which could in turn modulate the local effect size of a change in a state’s EITC matching rate (Chetty et al. 2013). To enhance my analysis and capture this feature, I recreate my key results while regressing on not just the state EITC matching rate and the ZIP code EITC participation rate (as before), but on the interaction of the two as well.

Theoretically, the effect of the matching rate by itself in a regression including its interaction with participation should be null. The interpretation here is how the outcome varies with matching rate when participation rate is zero, which should forestall any effect at all of changing rates. In contrast, the coefficient on the participation rate by itself should be negative, as there is always the federal EITC available to workers. More individuals participating in the EITC locally creates a similar logic to the model in Section II: more workers with work-contingent income will value employment more relative to unemployment, and be discouraged from reporting. Note that in Table 2, the coefficient on participation rate was negative and highly statistically significant in both the Poisson and probit specifications. Finally, the coefficient on the interaction term of matching and participation rates should be negative for the same reasons argued for the rate alone in the foregoing analysis without the interaction.

The results of this extended analysis are presented in Table 3. Taking the coefficients in turn, the state EITC matching rate alone displays a positive effect, significant at the 10% level. This is puzzling given the theoretical prediction of a null result, and may reflect unobserved heterogeneity between those ZIP codes with no EITC filers and those with some EITC filers - for example, they are likely to be higher-income and homogeneous compared to the average ZIP. There are only 750 ZIP-year observations in my dataset with no EITC filers, with an average of 0.05 reports across them. Ideally this value would be zero, but as my estimate is just a linear approximation, some deviation is tolerable.

The coefficient on the EITC participation rate is negative and highly statistically significant, aligning to the theoretical value. When more individuals in a locality participate in the EITC, the value of work relative to unemployment increases, and discourages reporting workplace crime such as wage theft. This result suggests that a state with no matching of the federal EITC would still see a decline in wage theft reports if workers increased their participation. The coefficient estimate of -0.622 implies that an increase in participation rate of 10 percentage points would decrease wage theft reports by about 6.2%.

Finally, the coefficient on the interaction term is large, negative, and highly statistically significant, confirming the theoretical prediction and validating the previous analysis in a more nuanced specification. Interpreting this value at the mean of state EITC participation rate (33%), the coefficient of -3.279 suggests that increasing the matching rate by 10 percentage points would decrease wage theft reporting by roughly 11% (ignoring for now the contribution of the coefficient on state EITC rate itself, which would tamper this effect slightly). While only an approximation, this would correspond to a state-level decrease of 66 fewer wage theft reports (off an average of 600 annually) and about \$726,000 in additional unreported stolen wages.

Table 3: Interaction Results

	(1) Complete Poisson	(2) Probit
State EITC Rate	0.680* (0.386)	0.314 (0.248)
EITC Participation Rate	-0.622*** (0.125)	-0.172* (0.089)
State EITC Rate $\times$ EITC Participation Rate	-3.279*** (1.097)	-1.596** (0.686)
State Unionization Rate	-0.636 (0.856)	-0.795 (0.659)
State Minimum Wage	0.001 (0.014)	-0.012 (0.010)
Governor Party	0.008 (0.025)	-0.007 (0.020)
Log Tax Returns	0.862*** (0.009)	0.636*** (0.005)
Share Below Poverty	1.481*** (0.108)	0.941*** (0.089)
Share Black	-0.135** (0.065)	0.105** (0.050)
Share <HS Education	-0.438*** (0.114)	-0.358*** (0.092)
Share Noncitizen	0.882*** (0.153)	1.644*** (0.128)
Share on Pub. Assist.	0.549*** (0.066)	0.130*** (0.044)
Share Est in Cons	-2.336*** (0.125)	-0.908*** (0.066)
Share Est in Ag	-1.413*** (0.400)	-0.071 (0.170)
Share Est in Food Svcs.	0.630*** (0.182)	0.917*** (0.102)
Share Emp <50	-7.510*** (0.551)	-5.534*** (0.207)
Log Population Density	-0.009 (0.007)	-0.001 (0.005)
Unemployment Rate	-1.368*** (0.345)	-0.797*** (0.217)
State & Year FE	✓	✓
R^2	0.34	0.33
N	218,947	218,947

Note: Standard errors are clustered at the state-year level. * = $p < 0.1$, ** = $p < 0.05$, *** = $p < 0.01$.

VI Secondary Outcomes Analysis

VI.A Secondary Outcomes - Empirical Approach

The results of the above estimation would not alone be sufficient to claim that there is a negative effect of transfers and reporting - it could be the case that transfers reduce theft frequency itself, mechanically generating a decline in reporting. To argue against this alternative mechanism, I analyze several outcomes which characterize employer behavior in wage theft. If the alternative mechanism is truly driving the observed change in reporting, one would expect to also see changes in these measures. I present each in turn for completeness.

Empty Reports

When a wage theft report is documented with no findings - that is, no backwages owed, no legal violations found, and no civil monetary penalties assessed - I refer to that as an “empty” report. While the exact nature of these is uncertain in the Wage & Hour Division data, it is reasonable to treat them as reports where no crime was identified. If employers were responding to transfers via an alternative mechanism and engaging in less wage theft, I would predict a positive effect of transfers on the count of empty reports, as there would be less wage theft to find. In contrast, if the effect of transfers on reporting is driven by the worker-side mechanism, then we would expect no change. I estimate the effect via similar model to Equation 6.

Total Violations

In a wage theft investigation, the authorities record each violation of the laws the Wage & Hour Division is responsible for enforcing. I offer this as a rough proxy for the severity / intensity of the wage theft taking place. In particular, if employers respond to transfer changes by reducing their wage theft, out of greater fear of being reported, it seems highly likely they would seek to reduce the number of laws they are breaking in the process. I use another Poisson model for estimating the effect of transfers on violations.

Civil Monetary Penalties

In addition to ordering repayment of backwages, wage theft investigations can lead to civil penalties for the employer in accordance with the laws they have violated. By similar thinking to penalties, this proxies for the intensive margin of wage theft, and should be decreasing if employers change their behavior. Unfortunately, this variable is not a count, but a continuous variable — specifically I use the log of the amount of penalties per employee involved in the case — so I cannot use a Poisson model.

Instead, I use a Heckman selection model to estimate the effect of transfers on civil penalties. The sample of reports wherein penalties are found are certainly a selected sample among wage theft reports, thus estimating a simple linear model would produce a biased estimate. For identification, I omit the following variables from the penalties estimation but include them in the binary step: Governor’s Party, State Minimum Wage, and share of the population earning over 400% of the poverty line. Each of these variables likely has an effect on whether or not penalties are assessed on a report, but do not influence in turn the amount of those penalties.

VI.B Secondary Outcomes - Results

I test my battery of secondary outcomes in 4. Interpreting these results (beginning in Column 1), the effect of EITC rates on the number of "empty" reports, where no wage theft or other workplace crime is found, is null. These empty reports are difficult to characterize - some may be agency-initiated investigations which fail to find evidence, or they could be "frivolous" reports filed by disgruntled employees. This difficulty notwithstanding, if it was true that employers were decreasing their wage theft behavior in response to rising EITCs, this would suggest a positive coefficient estimate, rather than the observed negative (though insignificant) estimate in Column 1.

Another null result is found for the effect on total labor violations in Column 2; however, one is not for the effect on civil monetary penalties in Column 3. While this result is less straightforward to interpret, it is certainly not the case that this result supports an employer reduction of intensity mechanism. The estimated effect is positive and highly statistically significant (even when bootstrapping in Appendix C). This suggests that more penalties are assessed on wage thefts reported at higher levels of state EITC, whereas a decrease in wage theft intensity should result in fewer penalties.

Returning to the model in Section II, this result agrees with the intuition presented. At higher EITC levels, workers are willing to "put up" with greater / more intense wage theft, as the value of work rises such that the risk of retaliatory firing is not worth running. Accordingly, those workers who do still choose to report are likely experiencing more severe wage theft, as for them the benefits of reporting still outweigh the potential costs. The results in Column 3 suggest that the wage theft reported at higher EITC levels is more severe / intense (more worthy of being penalized) than at lower EITC rates. A similar finding is reported in the context of sexual assault in Dahl and Knepper (2021), that the kinds of reports filed become more severe as outside options become less attractive relative to remaining employed.

Taken together, the results presented here for secondary outcomes of wage theft intensity rebuke the argument that the decreased reporting found in Sections IV and V is being driven

by changes in employer behavior. In particular, these results highlight a potential shift in the nature of the reports being filed, towards greater intensity, which is not only the opposite of a reduction in employer behavior, but is further in line with predictions of my model.

Table 4: Intensive Margin Results

	(1) Empty Reports Poisson	(2) Total Vltns. Poisson	(3) Civil Penalties Heckman
State EITC Rate	-0.163 (0.252)	0.131 (0.228)	-0.357 (0.475)
EITC Participation Rate	-0.374** (0.184)	-0.061 (0.137)	-0.415 (0.318)
State EITC Rate $\times$ EITC Participation Rate	0.293 (0.624)	-0.829 (0.566)	4.030*** (1.143)
State Unionization Rate	-1.422 (1.394)	0.299 (0.910)	0.843 (2.186)
State Minimum Wage	0.005 (0.020)	0.013 (0.014)	
Governor Party	-0.002 (0.036)	-0.053** (0.026)	
Log Tax Returns	0.255*** (0.013)	0.285*** (0.011)	-0.211*** (0.030)
Share Below Poverty	1.279*** (0.193)	0.687*** (0.169)	-0.682* (0.372)
Share Black	-0.294*** (0.084)	-0.154** (0.067)	0.364*** (0.140)
Share <HS Education	-0.676*** (0.175)	-0.465*** (0.145)	0.356 (0.370)
Share Noncitizen	0.966*** (0.170)	1.509*** (0.137)	0.403 (0.361)
Share on Pub Assist.	0.192* (0.108)	0.109 (0.091)	-0.374** (0.173)
Share Est in Cons	-0.399** (0.190)	-1.428*** (0.160)	0.502* (0.287)
Share Est in Ag	-0.202 (0.863)	0.032 (0.601)	0.525 (1.040)
Share Est in Food Svcs.	-0.772*** (0.274)	-0.166 (0.229)	0.168 (0.334)
Share Emp <50	-5.511*** (0.277)	-5.288*** (0.245)	2.862*** (0.613)
Log Population Density	-0.023** (0.010)	-0.010 (0.006)	-0.008 (0.023)
Unemployment Rate	-0.493 (0.480)	-0.765** (0.370)	0.313 (0.651)
State & Year FE	✓	✓	✓
N	66,421	66,421	66,421

Note: Standard errors are clustered at the state-year level. * = $p < 0.1$, ** = $p < 0.05$, *** = $p < 0.01$.

VII Conclusion

Wage theft is an understudied workplace crime, even though it poses a significant cost to American workers. Despite its prevalence, few workers choose to report to the government when they are the victims of wage theft, frustrating a deeper understanding of its dynamics. In this paper, I explore this underreporting through the established literature mechanism of retaliatory firing deterrence.

Workers faced with potential retaliatory firing for reporting workplace crime must weigh the value of employment against the costs of potential unemployment. As shown in recent works such as Dahl and Knepper (2021) and Johnson et al. (2023), when unemployment conditions worsen, workers are less likely to report, as the cost of reporting has effectively risen. In the foregoing analysis, I make two contributions to this literature. Firstly, I extend the mechanism of retaliatory firing into the specific context of wage theft, finding consistency with the results in other crimes.

Secondly, I augment the mechanism by altering not the cost of unemployment, but rather the value of employment itself, through work-contingent government transfers. Building this variable into the standard reporting model, I show how increased transfers should lower reporting propensities. Taking this model to the data, I confirm this result, showing that state EITC rates are negatively associated with wage theft reports. This finding is robust to a variety of empirical specifications.

I further defend this finding from the critique that a decrease in reports could reflect decreased wage theft activity, rather than a decrease in reporting propensities of workers. I show through several proxy measures that the intensity of wage theft does not have a negative relationship with state EITC rates, making it unlikely that employers are reducing their wage theft behavior in response to state matching rates. This makes the mechanism of reduced worker reporting the likely explanation for the observed phenomena. This finding improves the understanding of wage theft, and may inform policy designed to increase reporting propensities.

References

- Boone, J. and Ours, J. (2006). Are recessions good for workplace safety? *Journal of Health Economics*, 25:1069–93.
- Boone, J., van Ours, J. C., Wuellrich, J.-P., and Zweimüller, J. (2011). Recessions are bad for workplace safety. *Journal of Health Economics*, 30(4):764–773.
- Chetty, R., Friedman, J. N., and Saez, E. (2013). Using differences in knowledge across neighborhoods to uncover the impacts of the eitc on earnings. *American Economic Review*, 103(7):2683–2721.
- Clemens, J. (2021). How do firms respond to minimum wage increases? understanding the relevance of non-employment margins. *Journal of Economic Perspectives*, 35(1):51–72.
- Cooper, D. and Kroeger, T. (2017). Employers steal billions from workers’ paychecks each year. Technical report, Economic Policy Institute.
- Dahl, G. B. and Knepper, M. M. (2021). Why is workplace sexual harassment underreported? the value of outside options amid the threat of retaliation. Working Paper 29248, National Bureau of Economic Research.
- Galvin, D. J. (2016). Deterring wage theft: Alt-labor, state politics, and the policy determinants of minimum wage compliance. *Perspectives on Politics*, 14(2):324–350.
- Grittner, A. and Johnson, M. (2022). Deterring worker complaints worsens workplace safety: Evidence from immigration enforcement. SSRN Scholarly Paper ID 3943441, Social Science Research Network.
- Hertel-Fernandez, A. (2020). American workers’ experiences with power, information, and rights on the job: A roadmap for reform. Report, Roosevelt Institute.
- Internal Revenue Service (2022). Soi tax stats - individual income tax statistics - zip code data (soi).

- Ji, M. and Weil, D. (2010). Does ownership structure influence regulatory behavior? the impact of franchising on labor standards compliance. *Industrial and Labor Relations Review*, 68(5):977–1006.
- Johnson, M., Schwab, D., and Koval, P. (2023). Legal protection against retaliatory firing improves workplace safety. *Review of Economics and Statistics*.
- Kaplan, J. (2023). United States Governors 1775-2020. Inter-university Consortium for Political and Social Research.
- Kline, P. and Santos, A. (2012). A score based approach to wild bootstrap inference. *Journal of Econometric Methods*, 1(1):23–41.
- MacKinnon, J. G., Ørregaard Nielsen, M., and Webb, M. D. (2023). Cluster-robust inference: A guide to empirical practice. *Journal of Econometrics*, 232(2):272–299.
- Macpherson, D. and Hirsch, B. (2023). Five decades of cps wages, methods, and union-nonunion wage gaps at unionstats. com. *Industrial Relations: A Journal of Economy and Society*.
- Marinescu, I. and Rosenfeld, J. (2022). Worker power and economic mobility: A landscape report. Report, Urban Institute WorkRise report.
- Marinescu, I. E., Qiu, Y., and Sojourner, A. (2020). Wage inequality and labor rights violations. SSRN Scholarly Paper ID 3673495, Social Science Research Network, Rochester, NY.
- Meixell, B. and Eisenbrey, R. (2014). An epidemic of wage theft is costing workers hundreds of millions of dollars a year. Issue Brief 385, Economic Policy Institute.
- Melendez, R., Clarke, P., Khan, A., Gomez-Lopez, I., Li, M., and Chenoweth, M. (2023). National Neighborhood Data Archive (NaNDA): Socioeconomic Status and Demographic Characteristics of Census Tracts, United States, 2008–2020. Inter-university Consortium for Political and Social Research.

- Pararai, M. and Famoye, F. (2022). Generalized poisson-poisson mixture model for misreported counts with an application to smoking data. *Journal of Data Science*, 8(4):607–617.
- Roodman, D., Ørregaard Nielsen, M., MacKinnon, J. G., and Webb, M. D. (2019). Fast and wild: Bootstrap inference in stata using boottest. *The Stata Journal*, 19(1):4–60.
- Sainato, M. (2023). ‘I have not seen one cent’: billions stolen in wage theft from US workers”. *The Guardian*.
- Shapiro, I. (2019). State EITC provisions 1977-2018.
- Stansbury, A. (2021). Do us firms have an incentive to comply with the flsa and the nlra? Working Paper 21-9, Peterson Institute for International Economics.
- Tax Policy Center (2023). State eitc as percentage of the federal eitc.
- U.S. Census (2023a). American community survey 5-year data (2009-2021). Database, U.S. Department of Commerce.
- U.S. Census (2023b). County business patterns database. Database, U.S. Department of Commerce.
- US Department of Labor Wage & Hour Division (2023). Changes in basic minimum wages in non-farm employment under state law: Selected years 1968 to 2022.
- Weil, D. (2010). Improving workplace conditions through strategic enforcement. *Boston U. School of Management Research Paper*, (2010-20).
- Weil, D. (2018). Creating a strategic enforcement approach to address wage theft: One academic’s journey in organizational change. *Journal of Industrial Relations*, 60(3):437–460.
- Wooldridge, J. M. (1999). Distribution-free estimation of some nonlinear panel data models. *Journal of Econometrics*, 90(1):77–97.

Appendix

A Robustness - Dropping Top ZIPs by Wage Theft

One concern using the Wage & Hour Division data is that I cannot distinguish between worker-initiated and agency-initiated cases. The latter is not relevant to my analyses. To partially overcome this issue, I attempt to remove some of the cases which are most likely to be agency-initiated, specifically all those ZIP codes which experience at least 10 wage theft reports in any year. This should target those ZIP codes with the highest amount of wage theft, which are precisely those which are most often targeted for agency-initiated cases.

The Wage & Hour Division targets their investigations in areas with high density of wage theft and probable wage theft establishments, in order to maximize the deterrent effect of their investigations. That makes this drop likely to remove those ZIP codes where agency-initiated reports dominate the count of reports. In total, I drop about 0.10% of observations, accounting for about 17% of total wage theft cases in my sample. Given that the Wage & Hour Division reports roughly 25% per year, this seems likely to drop a high share of the agency-initiated cases, without significantly altering my dataset from the winsorized version.

Results from recreating the main analyses of this paper are presented below in Table 5. The key coefficients maintain similar size, magnitude, and significance levels, supporting the belief that my analysis would likely be robust to a sample of only worker-initiated reports. The exception is the result in Column 3, for the secondary outcome measure of empty reports count — the results for matching and participation rates are significant and negative, while the result for their interaction is significant and positive.

The combination of positive and negative effects here is puzzling — it is hard to conceive of a mechanism by which increased EITC matching and participation rates leads to less empty reports, while their interaction would have a positive association. This could mean that the outcome of empty report counts is simply not a reliable measure of employer wage theft intensity, and is generating spurious results. Even if not, it is difficult to rationalize this result with a decrease in employer wage theft intensity, given the contradictory results.

Table 5: Robustness - Dropping Top ZIPs

	Reports		Secondary Outcomes		
	(1) Complete Poisson	(2) Probit	(3) Empty Reports Poisson	(4) Total Violations Poisson	(5) Civil Penalties Heckman
State EITC Rate	0.680* (0.374)	0.302 (0.236)	-0.481** (0.239)	0.008 (0.204)	-0.323 (0.461)
EITC Participation Rate	-0.486*** (0.115)	-0.141 (0.086)	-0.315** (0.160)	0.083 (0.137)	-0.373 (0.313)
State EITC Rate × EITC Participation Rate	-3.380*** (1.032)	-1.556** (0.650)	1.182** (0.503)	-0.500 (0.493)	4.012*** (1.140)
State Unionization Rate	-0.820 (0.911)	-0.795 (0.662)	-1.498 (1.453)	0.114 (0.906)	0.116 (2.229)
State Minimum Wage	0.003 (0.016)	-0.013 (0.010)	0.001 (0.020)	0.007 (0.014)	
Governor Party	0.008 (0.026)	-0.007 (0.020)	-0.002 (0.039)	-0.049* (0.026)	
Log Tax Returns	0.874*** (0.009)	0.631*** (0.006)	0.217*** (0.014)	0.257*** (0.010)	-0.220*** (0.027)
Share Below Poverty	1.407*** (0.112)	0.884*** (0.090)	1.037*** (0.198)	0.306* (0.168)	-0.522 (0.414)
Share Black	-0.110** (0.055)	0.119** (0.047)	-0.186*** (0.068)	-0.085 (0.057)	0.304** (0.143)
Share < Education	-0.680*** (0.110)	-0.372*** (0.091)	-0.883*** (0.171)	-0.494*** (0.134)	0.326 (0.378)
Share Noncitizen	0.970*** (0.138)	1.549*** (0.128)	0.774*** (0.201)	1.256*** (0.145)	0.165 (0.365)
Share on Pub Assist.	0.533*** (0.064)	0.126*** (0.044)	0.112 (0.111)	0.087 (0.086)	-0.425** (0.176)
Share Est in Cons	-2.065*** (0.114)	-0.872*** (0.065)	0.025 (0.169)	-1.031*** (0.140)	0.388 (0.280)
Share Est in Ag	-1.151*** (0.405)	-0.056 (0.170)	0.117 (0.798)	0.121 (0.570)	0.308 (0.962)
Share Est in Food Svcs.	0.760*** (0.163)	0.906*** (0.098)	-0.601** (0.254)	0.000 (0.201)	0.006 (0.344)
Share Emp <50	-7.544*** (0.525)	-5.372*** (0.204)	-4.923*** (0.265)	-4.715*** (0.240)	2.726*** (0.611)
Log Population Density	-0.011* (0.006)	-0.002 (0.005)	-0.013 (0.009)	-0.003 (0.006)	0.009 (0.021)
Unemployment Rate	-1.367*** (0.331)	-0.777*** (0.215)	-0.288 (0.473)	-0.581* (0.341)	0.258 (0.665)
State & Year FE	✓	✓	✓	✓	✓
N	218,947	218,947	66,421	66,421	66,421

Note: Standard errors are clustered at the state-year level. * = $p < 0.1$, ** = $p < 0.05$, *** = $p < 0.01$.

B EITC Rate Definitions

The values used for state EITC rates are discussed in Section III but bear further explanation. While many states use a simple matching rate, several use alternative structures. Here I review those alternatives and how I encoded values from the description:

California

California's credit has a smaller eligible income range than the federal credit, with a maximum which rose over the active years in my sample of analysis. The Tax Policy Center has summarized this divergence in recent years by listing the rate as a 45% match (rather than the statutory 85% figure). I follow this example and use the 45% rate in my set.

Maryland

Maryland employs two separate matching rates, one for refundable credits (used in the baseline) and one for non-refundable at 50%. To approximate the effective rate of this combination, I use the average of the varying statutory rate and 50%.

Minnesota

Minnesota's credit is different from other states in that it is not structured as a percentage of the federal credit. The Tax Policy Center reports an average rate of 33%, which I employ in the baseline. I do not adjust this rate for my preferred set, but simply report this usage for completeness.

New Jersey

New Jersey limits their match to a maximum income of \$20,000. This is roughly half the total federal EITC range, though it is in the "plateau" region of the maximum credit. Accordingly, as an alternative to the statutory rate, I use three-quarters of that rate to reflect the diminished significance.

New York

New York City has its own supplemental 5% EITC. Given that roughly 40% of the state lives in the city, I adjust the statutory rate of 30% up to 32% in my preferred set.

Ohio

Ohio's credit is limited to 50% of liability for state taxable income above \$20,000. I halve Ohio's rates to attempt and reflect this diminished significance.

Oregon

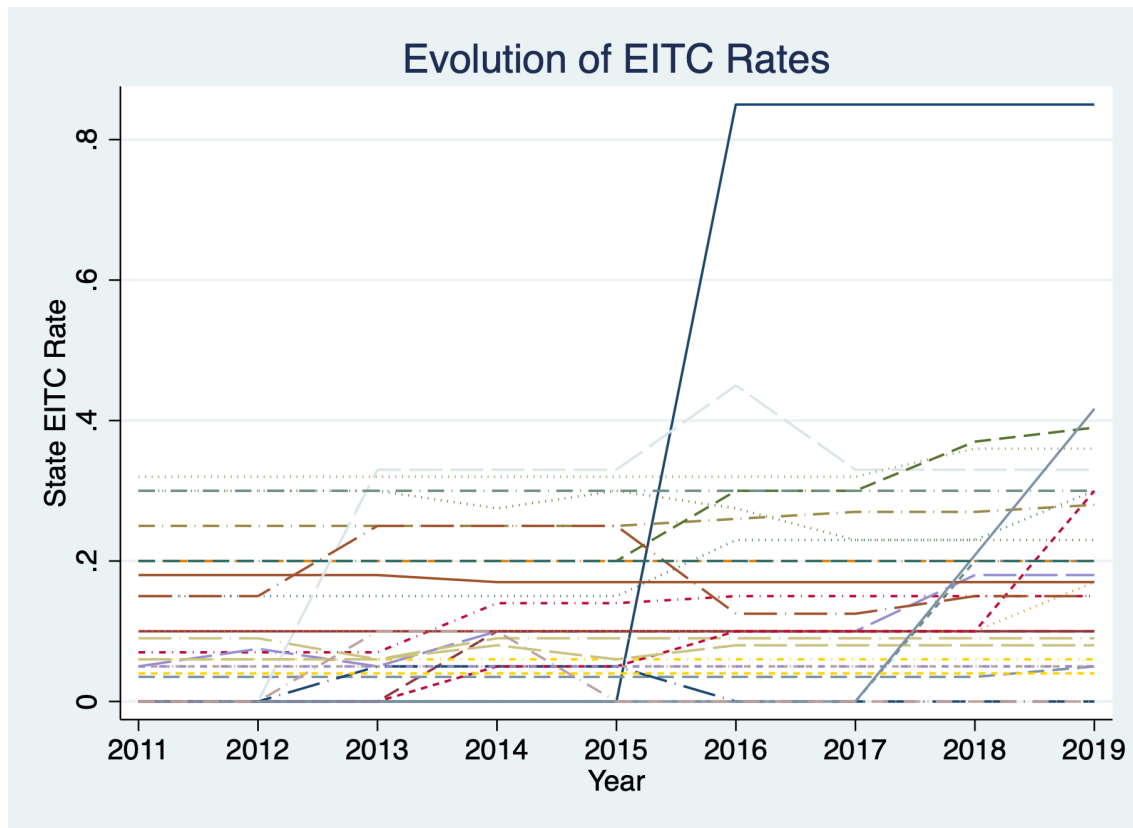
Oregon switched from a single rate for all filers to having a separate rate for filers with children under the age of three in 2017. Accordingly, I adjust the statutory rate of 8% up to 9% as my preferred rate, to reflect that a share of filers are receiving the new separate higher rate of 11%.

Wisconsin

Wisconsin's credit is only available to workers with qualified children, and uses separate rates by number of children, as one (4%), two (11%), and three (34%). To try and match this in an aggregate, I use the 4% figure in the statutory and adjust to 10% in the preferred rate.

Finally, the evolution of the statutory rates can be visualized below in Figure 3. While labelling every state is impractical, in anticipation of interest I note that the state which transitioned from no rate to 85% match in 2016 is California.

Figure 3: Evolution of Statutory Rates



C Cluster Size Heterogeneity

While clustering at the state-year level ensures that I have a relatively large number of clusters and allows for what I believe is the appropriate level of error correlation, there is significant size heterogeneity among those clusters. This is due to the variation in number of underlying ZIP codes per state, which vary by an order of magnitude at the extremes (despite most being fairly similar in the several hundreds count). This kind of heterogeneity can lead to over-rejection in hypothesis testing, as the larger clusters will tend to have much higher leverage, meaning that asymptotic theory may not carry through reliably. To overcome this issue, I employ the wild score cluster bootstrap approach, as developed by Kline and Santos (2012) and recommended in this setting by MacKinnon et al. (2023).

Implementing this via the Stata command `boottest` (Roodman et al. 2019), I recreate the key portions of Tables 3 and 4 below with the significance levels calculated accordingly. The results are largely unchanged, giving confidence that the cluster size heterogeneity is not driving inference misleadingly.

Table 6: Main Results - Bootstrapped Significance

	Reports		Secondary Outcomes		
	(1)	(2)	(3)	(4)	(5)
	Complete Poisson	Probit	Empty Reports Poisson	Total Violations Poisson	Civil Penalties Heckman
State EITC Rate	0.680*	0.314	-0.163	0.131	-0.357
Bootstrap P-value	(0.056)	(0.176)	(0.804)	(0.558)	(0.002)
EITC Participation Rate	-0.622***	-0.172**	-0.374**	-0.061	-0.415
Bootstrap P-value	(0.000)	(0.048)	(0.257)	(0.653)	(0.016)
State EITC Rate × EITC Participation Rate	-3.279***	-1.596**	0.293	-0.829	4.030***
Bootstrap P-value	(0.002)	(0.012)	(0.967)	(0.136)	(0.000)
N	218,947	218,947	66,421	66,421	66,421

Note: P-values are presented in parentheses, and were calculated using the wild score cluster bootstrap.

* = $p < 0.1$, ** = $p < 0.05$, *** = $p < 0.01$.